

On the role of bimolecular reactions in chemical activation systems

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The kinetics of chemically activated intermediates in complex-forming bimolecular reactions can often be described in terms of the branching ratio between their unimolecular decomposition channels and the collisional stabilization under steady-state conditions (D/S). This requires that the stabilized part of the population does not contribute to the decomposition rate, which is true only for reaction times shorter than the thermal lifetime of the intermediate. As these intermediates, however, are often radicals or other highly reactive species, they can undergo consecutive bimolecular reactions. Here two limiting cases are conceivable: (i) If the bimolecular steps are slow, mainly stabilized intermediates react, and their population is prevented from being built up, *i.e.* their thermal decomposition is suppressed. In this case the steady-state description in terms of D/S remains adequate also for reaction times longer than the thermal lifetime of the intermediate. (ii) If the bimolecular reactions are extremely fast, even highly excited intermediates react and are removed from the population. Consequently, the observed relative yield of the unimolecular channels, Φ , drops below the steady-state value $\Phi_{ss} = [1 + (D/S)^{-1}]^{-1}$. The branching between decomposition and stabilization of a chemically activated intermediate, therefore, can depend on the rate of its bimolecular consecutive reactions, a fact which is often overlooked in the kinetic analysis of such systems. In the present work we investigate the dependence of Φ on the rate of these bimolecular steps in a general way by means of a master equation. A simple method is presented to estimate the parameter range, where the steady-state approach in terms of D/S remains adequate if bimolecular reactions of the intermediate occur. The problem is exemplarily treated for the reaction between chemically activated but-2-yl radicals and hydrogen atoms.

1. Introduction

Complex-forming bimolecular reactions proceeding *via* a chemically activated adduct play a crucial role in combustion^{1–3} and atmospheric chemistry.^{4–6} Their kinetics under steady-state conditions is often characterized in terms of the competition between unimolecular decomposition and collisional stabilization of the excited intermediate.^{7–9} Convenient measures are the ratio of the decomposition over stabilization rate, D/S , or the relative branching fraction of the unimolecular decomposition channels, $\Phi_{ss} = D/(D + S) = [1 + (D/S)^{-1}]^{-1}$.^{7,9}

At short reaction times the kinetics of product formation is governed by the highly excited initial population of the adduct and the corresponding specific rate coefficients of its decomposition. At longer times the growing fraction of collisionally deactivated species becomes increasingly important, and finally the largest eigenvalue of the J -matrix,⁸ *i.e.* the thermal unimolecular rate coefficient, determines the rate of product formation. This time dependence of the branching between unimolecular decomposition and collisional stabilization under chemical activation conditions was first discussed by Snider¹⁰ and by Pritchard and Vatsya¹¹ and investigated in more detail by Schranz and Nordholm.¹² General expressions for the temporal behaviour of the molecular populations in such systems were derived by Smith, McEwan, and Gilbert.^{8,13}

In applications the fate of the intermediates after collisional deactivation is normally not traced further despite these adducts being mostly radicals or other highly reactive species. If, however, the stabilized intermediates do not react with other species in bimolecular steps, they will finally undergo the same unimolecular reactions as the excited species and,

given a sufficiently long reaction time, no stabilized adduct will survive; the sum over all relative product yields of the unimolecular reaction channels will be unity.

In contrast to collisional deactivation a bimolecular reaction causes an irreversible removal of the intermediates. In an earlier work¹⁴ we proposed to account for this effect by including a bimolecular sink term in the master equation. The consequences for the overall kinetics of chemical activation systems, however, have not yet been investigated in detail. Hence in the present work we address the question as to what extent observed product yields can be affected by the existence of such a bimolecular step.

The practical relevance of this problem can be illustrated by two examples from atmospheric chemistry. Reactions of olefins with NO_3 form oxiranes *via* the unimolecular decomposition of the intermediate adduct radicals.⁴ In air the relative oxirane yields decrease with increasing pressure, whereas pressure independent yields of unity are often observed in inert bath gases.¹⁵ This finding can be explained by stressing the above arguments. In air only excited intermediates contribute to the oxirane formation, because the stabilized radicals are effectively removed by reaction with O_2 . The decreasing oxirane yield with increasing pressure simply reflects the pressure-dependent collisional stabilization. In an inert bath gas, however, the deactivated species remain a part of the population and finally also decompose to yield oxirane. Accordingly, pressure independent yields near unity are obtained if the time-scale of the experiment is larger than the thermal lifetime of the adduct radical. Similar arguments are likely to apply to the fate of the Criegee-intermediates occurring in the gas-phase ozonolysis of olefins.⁴ If the stabilized Criegee-radicals are not scavenged in a bimolecular reaction (the nature of which

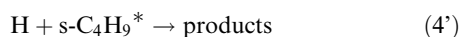
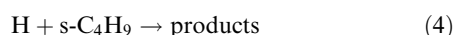
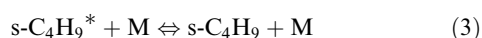
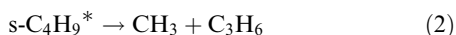
under atmospheric conditions is not yet known), the relative yield of the decomposition products is unity.^{16,17}

The most general description of a chemical activation system is accomplished by solving the appropriate time-dependent master equation. This is necessary, in particular, at elevated temperatures where the rates of chemically and thermally activated reactions of the intermediate become comparable.¹⁸ At low temperatures, however, the steady-state approximation is a much simpler and often very useful approach.¹³ We therefore concentrate in the present work on the influence of the bimolecular reactions on the steady-state population. As only the basic approach should be demonstrated and the most important general consequences are to be discussed, we employ the well-characterized^{7,9,19,20} chemically activated decomposition of s-C₄H₉ radicals as an example. The application to other reaction systems is straightforward if the necessary input data are available.

2. Modeling calculations

2.1. The master equation

We consider the reaction of *cis*-C₄H₈ with a large excess of H atoms. In this case the s-C₄H₉ radicals formed, besides unimolecular dissociation and collisional deactivation, can undergo a consecutive reaction with H to form several products:



Here M denotes an inert collision partner and the symbol * stands for rovibrational excitation. Reaction (−1) can be neglected because reaction (2) is much faster due to its lower threshold energy (see below).

The master equation characterizing the s-C₄H₉ population, $n_i = n(E_i)$, reads as^{7–9}

$$\frac{dn_i}{dt} = R_1 f_i - \omega n_i + \omega \sum_j P_{ij} n_j - k_{2i} n_i - k_4 [\text{H}] n_i \quad (5)$$

where $f_i = f(E_i)$ denotes the normalized s-C₄H₉ distribution as generated by reaction (1) and $R_1 = k_1 [\text{H}] [\text{cis-C}_4\text{H}_8]$ the corresponding rate of reaction. The second and third terms characterize the collisional depopulation and population, respectively, of the energy level i with ω being the collision frequency and $P_{ij} = P(E_i \leftarrow E_j)$ the probability of a collision-induced transition $E_i \leftarrow E_j$. The term $k_{2i} n_i$ represents the rate of the unimolecular decomposition step (2) with the specific rate coefficient $k_{2i} = k_2(E_i)$. Finally, the last term describes the bimolecular reactions (4) and (4'), respectively, where it is assumed that k_4 does not depend on the excitation energy of s-C₄H₉ i.e. $k_4 = k_{4'}$. This assumption can be easily dropped if $k_4(E)$ is known. The factor $k_4 [\text{H}]$ is a pseudo-first order rate coefficient because $[\text{H}] \gg [\text{cis-C}_4\text{H}_8]$. Eqn. (5) can be cast into matrix form^{7–9} and by assuming steady-state conditions, $dn_i/dt = 0$, one obtains

$$R_1 \mathbf{F} = \{\omega(\mathbf{I} - \mathbf{P}) + \mathbf{K}_2 + k_4 [\text{H}] \mathbf{I}\} \mathbf{N}^s \equiv \mathbf{J} \mathbf{N}^s \quad (6)$$

Here the vector/matrix symbols correspond to the symbols in eqn. (5) and $\mathbf{I}$ denotes the unit matrix. The steady-state population $\mathbf{N}^s$ then follows by inversion of the matrix $\mathbf{J}$ as

$$\mathbf{N}^s = R_1 \mathbf{J}^{-1} \mathbf{F} \quad (7)$$

and can be normalized using the relation

$$\tilde{\mathbf{N}}^s = \frac{\mathbf{J}^{-1} \mathbf{F}}{\sum_i (\mathbf{J}^{-1} \mathbf{F})_i} \quad (8)$$

(here and in the following, notations like $(\mathbf{X})_i$ are used for the diagonal elements of a matrix $\mathbf{X}$). The rate of the decomposition reaction (2) is now obtained by averaging $k_2(E)$ over the steady-state distribution $\tilde{n}^s(E)$:

$$R_2 = \sum_i (\mathbf{K}_2 \tilde{\mathbf{N}}^s)_i \quad (9)$$

and for the relative branching fraction of channel (2) it follows from eqns. (7) and (9):

$$\Phi_2 \equiv \frac{R_2}{R_1} = \sum_i (\mathbf{K}_2 \mathbf{J}^{-1} \mathbf{F})_i \quad (10)$$

By fixing all the other parameters, the dependence $\Phi_2 = \Phi_2(k_4 [\text{H}])$ can be examined.

In our actual computations the master equation is set up with a grain size of 10 cm^{−1}, and the energy is counted from the vibrational ground state of s-C₄H₉. We calculate the specific rate coefficients and molecular distributions for an angular momentum quantum number $J = \langle J \rangle_{300 \text{ K}} = 40$, corresponding to the average rotational excitation of s-C₄H₉ at 300 K. That is, we assume the thermal rotational distribution of s-C₄H₉ to be not disturbed by the reactive processes. This is evident for reactions (1) and (2), which proceed *via* tight transition states,^{7–9} and is surely a reasonable approximation in the case of reactions (4) and (4'), because $k_4 [\text{H}]$ cannot normally exceed the collision frequency.

In the following sections the different quantities entering eqns. (6)–(10) are briefly discussed.

2.2. The nascent distribution $f(E)$

The nascent population of s-C₄H₉ as generated by reaction (1) is approximated by a shifted thermal distribution^{7–9}

$$f(E) = N W_{\text{s-C}_4\text{H}_9}(E - E_{0(-)}) \exp[-(E - E_{0(-)})/k_B T] \quad (11)$$

where N is a normalization factor and $W_{\text{s-C}_4\text{H}_9}$ the sum of states of the transition state of reaction (1) approximated by the sum of states of s-C₄H₉;⁷ k_B denotes Boltzmann's constant, and T the temperature. The minimum energy of this distribution is equal to the threshold energy of the reverse reaction of eqn. (1), $E_{0(-)}$, and from the thermochemical data²¹ for H, *cis*-C₄H₈, and s-C₄H₉ with a threshold energy of 8.5 kJ mol^{−1} for reaction (1),^{22,23} a value of $E_{0(-)} \approx 152$ kJ mol^{−1} (12 700 cm^{−1}) is obtained. The resulting distribution $f(E)$ is shown in the top panel of Fig. 1.

2.3. The specific rate coefficients $k_2(E)$

The specific rate coefficients $k_2(E)$ are obtained from RRKM theory:^{7–9,24}

$$k_2(E) = \frac{L^\ddagger W_{\text{TS2}}(E - E_{0(2)})}{h \rho_{\text{s-C}_4\text{H}_9}(E)} \quad (12)$$

where $L^\ddagger = 2$ denotes the reaction-path degeneracy,^{7–9} W_{TS2} the sum of states of the transition state of reaction (2), $\rho_{\text{s-C}_4\text{H}_9}$ the density of states of s-C₄H₉, h Planck's constant, and $E_{0(2)}$ the threshold energy of reaction (2). For $E_{0(2)}$ a value of 130.0 kJ mol^{−1} (10 870 cm^{−1}) was adopted,^{26,27} and the sums and densities of states were exactly counted^{8,9,28} (for $J = 40$, see above) with molecular data from refs. 27 and 29. The resulting specific rate coefficients are also displayed in the top panel of Fig. 1.

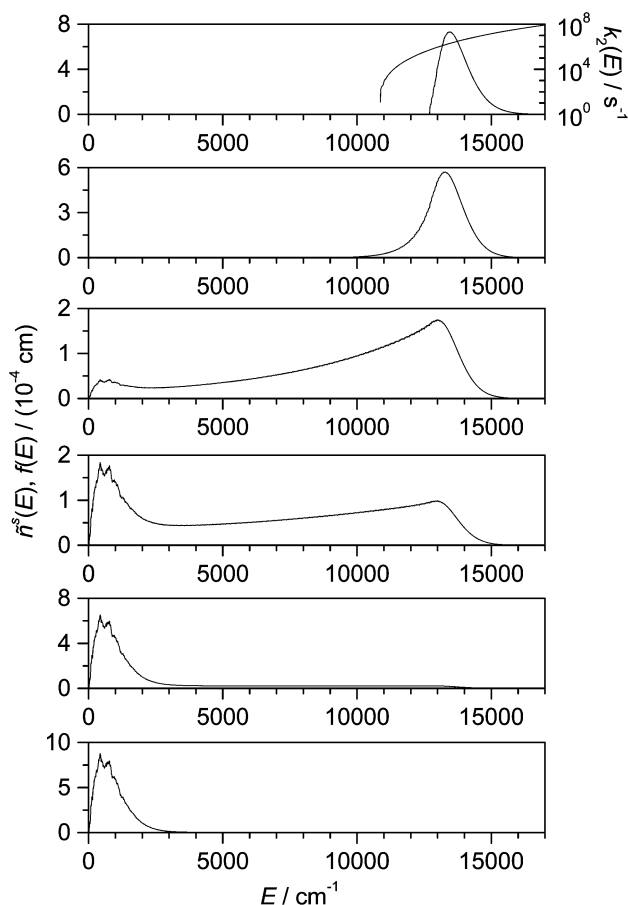


Fig. 1 Normalized distribution functions of s-C₄H₉. Top panel: $f(E)$ and $k_2(E)$; 2nd to 6th panel: $\tilde{n}^s(E)$ for $k_4[\text{H}] = 1 \times 10^7 \text{ s}^{-1}$, $1 \times 10^6 \text{ s}^{-1}$, $5 \times 10^5 \text{ s}^{-1}$, $1 \times 10^5 \text{ s}^{-1}$ and $1 \times 10^4 \text{ s}^{-1}$, respectively; for conditions see text. The distribution functions for $k_4[\text{H}] \leq 1 \times 10^4 \text{ s}^{-1}$ are visually indistinguishable.

2.4. The transition probabilities $P(E_i, E_j)$

As only the general effects of including a bimolecular reaction into a steady-state master equation are to be examined, a stepladder model^{7–9} for the transition probabilities is sufficient. In this simple approach merely transitions between states separated by a fixed amount of energy, ΔE_{SL} , are allowed.

For this model the two general constraints for transition probabilities, completeness and detailed balancing,^{8,9} read as

$$P(E_{j-1}, E_j) + P(E_{j+1}, E_j) = 1 \quad (13)$$

and

$$\frac{P(E_{j+1}, E_j)}{P(E_j, E_{j+1})} = \frac{\rho_{\text{s-C}_4\text{H}_9}(E_{j+1})}{\rho_{\text{s-C}_4\text{H}_9}(E_j)} \exp\left(-\frac{\Delta E_{\text{SL}}}{k_B T}\right) \quad (14)$$

respectively.

The only parameter, ΔE_{SL} , can be related to the average energy transferred per collision, $\langle \Delta E \rangle$, by^{30,31}

$$\langle \Delta E \rangle \approx -\Delta E_{\text{SL}} \tanh\left(\frac{\Delta E_{\text{SL}}}{2F_E k_B T}\right) \quad (15)$$

where the quantity F_E characterizes the energy dependence of the density of states.³² By employing the molecular data for s-C₄H₉ from ref. 29, we obtain $F_E \approx 1.33$ for $T = 300 \text{ K}$. Furthermore, a value of $\Delta E_{\text{SL}} = 190 \text{ cm}^{-1}$ is used²⁵ which corresponds to $\langle \Delta E \rangle \approx -60 \text{ cm}^{-1}$. One should note that the energetic resolution of the calculated molecular distribution functions is governed by the grain size of the master equation (10 cm^{-1}) and not by the step size ΔE_{SL} . This is achieved by actually splitting the master equation into a set of $190 \text{ cm}^{-1}/$

$10 \text{ cm}^{-1} = 19$ subequations, which are energetically shifted with respect to each other by 10 cm^{-1} and which are solved separately. In this way the poor resolution of a stepladder model with 190 cm^{-1} step size is accordingly improved, and the averaging of the rate coefficients becomes much more precise.

In order to facilitate the discussion below, it is useful to express the rate of stabilization in terms of an effective collision frequency, $\gamma_c \omega$, with a collision efficiency γ_c following from ref. 33:

$$\frac{\gamma_c}{1-\gamma_c^2} = \frac{-\langle \Delta E \rangle s^*}{\langle E \rangle - E_{0(2)}} \quad (16)$$

Here $\langle E \rangle$ is the average energy of the nascent s-C₄H₉ distribution, $f(E)$, and the parameter s^* characterizes the energy dependence $k_2(E)$.³³ With $\langle E \rangle = 13770 \text{ cm}^{-1}$ and $s^* = 4.8$ one obtains $\gamma_c \approx 0.1$.

The stepladder model causes the matrix **J** to be tridiagonal, and eqn. (10) can be solved by standard procedures (routines bandec and banbks from ref. 34). It should be noted, however, that the quantities ω , $k_2(E)$, and $k_4[\text{H}]$, the sum of which is needed in eqn. (6), differ by up to 20 orders of magnitude (see below), and it is essential to use compiler options that enable 128 bit word length.

3. Results and discussion

Before considering the detailed results of the master-equation modeling, it is useful to visualize the fundamental effects in terms of elementary kinetic considerations based on eqns. (1)–(4').

At first, as in the usual treatment of a chemically activated reaction, we neglect the bimolecular reactions (4) and (4'). In this case, after starting reaction (1) at $t = 0$, a steady-state population of s-C₄H₉* is established after a short incubation period,¹³ and the excited radicals formed are subject to the competition between decomposition (2) and collisional stabilization (3). The concentration of stabilized s-C₄H₉ increases with time^{8,12,13} but as long as it is sufficiently low, collisional activation, eqn. (-3), is unimportant. Then from $d[\text{s-C}_4\text{H}_9^*]/dt = 0$, it follows for the rate of reaction: $R_1 = R_2 + R_3 - R_{-3} \approx R_2 + R_3$. This regime is referred to in ref. 12 as the *intermediate* steady-state. In this time interval, the traditional chemical activation formalism applies, *i.e.* the branching ratio between decomposition and net stabilization, $R_2/(R_3 - R_{-3}) \approx R_2/R_3$ (often termed D/S),^{7,9} is virtually independent of time and represents an adequate description of the reaction system.¹² For the relative yield of reaction (2) it follows that $\Phi_2 \equiv R_2/R_1 = R_2/(R_2 + R_3) < 1$. In the respective steady-state master equation the quasi-irreversible stabilization has to be accounted for by either neglecting upward collisions⁷ or, if detailed balancing is included, by introducing a lower absorbing barrier in the collisional deactivation cascade at energies below the lowest reaction threshold.⁹

After a sufficiently long reaction time (of the order of $0.1/k_2^{\text{uni}}$, where k_2^{uni} is the thermal rate coefficient of reaction (2) at the actual temperature and pressure),¹² the stabilization reservoir is filled up and the rates of collisional deactivation and activation become comparable, *i.e.* $R_3 \approx R_{-3}$. This is the *final* steady-state¹² (see also refs. 8, 10, 11 and 13) where the ratio of decomposition over net stabilization, $R_2/(R_3 - R_{-3})$, loses its physical sense, and one trivially has $R_1 = R_2$ and hence $\Phi_2 = 1$. It should be emphasized that a steady-state master equation with upward collisions included corresponds to this case if no absorbing barrier exists.

The situation changes considerably if the bimolecular reactions (4) and (4') are taken into account. These reactions form a physically reasonable sink for the intermediates, and the

artificial condition of a lower absorbing barrier in the master equation must be dropped. If the absorbing barrier would be retained, an irreversible loss of s-C₄H₉ from the reaction sequence eqns. (1)–(4') would result and as a consequence a too low product yield would be predicted.

Because generally both excited and stabilized but-2-yl radicals react with H atoms, their steady-state distribution and hence their unimolecular kinetics depend on the rate of this reaction. If the bimolecular reaction is faster than the collisional deactivation, excited intermediates, s-C₄H₉^{*}, react and stabilization becomes unimportant. If, however, the collisional deactivation dominates, mainly stabilized s-C₄H₉ radicals will react with H atoms. It is presumed that reaction (4) is sufficiently fast so as to remove the stabilized but-2-yl radicals before reactivation; this corresponds to an irreversible stabilization and a regime is maintained equivalent to the intermediate steady-state mentioned above. If the rate of reaction (4) is much lower than the rate of collisional activation (-3), a situation corresponding to the final steady-state arises, *i.e.* the yield of reaction (2) approaches unity.

In the following we quantify these considerations in terms of eqns. (8) and (10). We assume conditions typical for a low-pressure chemical activation experiment: $T = 300$ K, $P = 5$ mbar, and He as bath gas. A Lennard–Jones collision number $Z_{LJ}(s\text{-C}_4\text{H}_9/\text{He}) \approx Z_{LJ}(n\text{-C}_4\text{H}_{10}/\text{He}) = 1.42 \times 10^7 \text{ mbar}^{-1} \text{ s}^{-1}$ is used³⁵ which results in an effective collision frequency $\gamma_c \omega = \gamma_c Z_{LJ} P = 7.1 \times 10^6 \text{ s}^{-1}$. The pseudo-first order rate coefficient $k_4[\text{H}]$ is treated as a parameter and varied between $1 \times 10^8 \text{ s}^{-1}$ and $1 \times 10^{-14} \text{ s}^{-1}$ which formally corresponds to H-atom concentrations between $2 \times 10^{18} \text{ cm}^{-3}$ and $2 \times 10^{-4} \text{ cm}^{-3}$ ($k_4 = 5.0 \times 10^{-11} \text{ cm}^3 \text{ s}^{-1}$).²³ We note that in conventional discharge flow-reactors H-atom concentrations typically lie in the range 10^{12} – 10^{14} cm^{-3} .³⁶

The normalized steady-state distributions from eqn. (8) for these conditions are displayed in Fig. 1. It can be easily envisaged that with a decreasing rate of the bimolecular reaction the steady-state population of s-C₄H₉ gradually changes from the nascent chemically activated distribution to the thermal distribution with a broad transition range near $k_4[\text{H}] \approx 0.1\gamma_c\omega$. An analogous picture arises if $k_4[\text{H}]$ is held fixed and the pressure is increased.

The corresponding relative branching fractions for reaction (2) calculated from eqn. (10) are displayed in Fig. 2 and selected results are compiled in Table 1. For $1 \times 10^{-7} \text{ s}^{-1} < k_4[\text{H}] < 1 \times 10^5 \text{ s}^{-1}$ a value of $\Phi_2 \approx 0.43$ is obtained which agrees with the result from a master-equation analysis neglecting reactions (4) and (4') and using an absorbing barrier instead. From the plateau in Fig. 2 it becomes evident that this

Table 1 Relative branching fractions of reaction (2) with respect to the pseudo-first order rate coefficient of the consecutive bimolecular reaction

$k_4[\text{H}]/\text{s}^{-1}$	Φ_2
1×10^7	0.2015
1×10^6	0.3830
5×10^5	0.4050
1×10^5	0.4247
1×10^4	0.4295
1×10^3	0.4300
1×10^{-4}	0.4300
1×10^{-5}	0.4301
1×10^{-6}	0.4303
1×10^{-7}	0.4324
1×10^{-8}	0.4526

traditional treatment provides an adequate description of the system over a broad range of conditions; it fails, however, if the rate of the bimolecular reaction becomes comparable either to the rate of collisional deactivation or to the rate of the thermal decomposition of the intermediate as was already discussed above in a qualitative manner.

In the first case, $k_4[\text{H}] \approx \gamma_c\omega$, the yield Φ_2 falls below the plateau value because excited but-2-yl radicals react. From Fig. 2 it becomes evident that the deviations stay below 10% as long as $k_4[\text{H}] < 0.1\gamma_c\omega$. For mono and diatomic bath gases at room temperature the collision efficiency γ_c is often of the order of 0.1 and accordingly the relation $k_4[\text{H}] < 0.01\omega$ represents a rough estimate for the upper limit of $k_4[\text{H}]$ below which the intermediate steady-state is essentially undisturbed.

In the second case, $k_4[\text{H}] \approx k_2^{\text{uni}}$, the thermal decomposition of s-C₄H₉ can effectively compete with the bimolecular step. Consequently, the yield Φ_2 is increased and reaches unity if the thermal unimolecular reaction prevails. For our actual conditions we obtain $k_2^{\text{uni}} = 4.1 \times 10^{-10} \text{ s}^{-1}$ as the lowest eigenvalue of the matrix **J** (routine tqll from ref. 34 applied after symmetrization of **J**,⁸ note that due to our definition of **J**, eqn. (6), the eigenvalues are positive). The deviations of Φ_2 from the intermediate steady-state value remain below 10% for $k_4[\text{H}] > 10k_2^{\text{uni}}$ as is again evident from Fig. 2.

We conclude that within the range $0.01\omega > k_4[\text{H}] > 10k_2^{\text{uni}}$ the observed product yield Φ_2 is obviously an adequate measure of the branching between the chemically activated decomposition and the collisional stabilization of the excited but-2-yl radicals. It is instructive to compare this interval with the results from the time-dependent approach in ref. 12 (chemical activation without a consecutive bimolecular reaction). The authors give an approximate time range of $(0.1\lambda_F)^{-1} < t < (10k^{\text{uni}})^{-1}$ for the existence of the intermediate steady-state where k^{uni} is the thermal rate coefficient of the unimolecular step and λ_F the eigenvalue of the **J** matrix that corresponds to the eigenvector most closely resembling the initial distribution $f(E)$. For our conditions we obtain $\lambda_F \approx 1.3 \times 10^8 \text{ s}^{-1} \approx 2\omega$ and hence $(0.2\omega)^{-1} < t < (10k_2^{\text{uni}})^{-1}$ for the time range of the intermediate steady-state. The approximate correspondence of this time range with the above rate-coefficient interval is evident. Obviously, the quantity $(k_4[\text{H}])^{-1}$ can be loosely interpreted as an average reaction time for the chemically activated population of s-C₄H₉ within the reaction sequence of eqns. (1)–(4'), a fact that seems reasonable. The difference between $(0.01\omega)^{-1}$ and $(0.2\omega)^{-1}$ at the lower end of the time scale may be due to interferences between the collisional relaxation process and the bimolecular reaction. Here a more quantitative explanation would call for a solution of the time-dependent master eqn. (5). This, however, is not the intention of the present work.

The behaviour exemplarily demonstrated for s-C₄H₉ is general for all chemical activation systems, and with the above

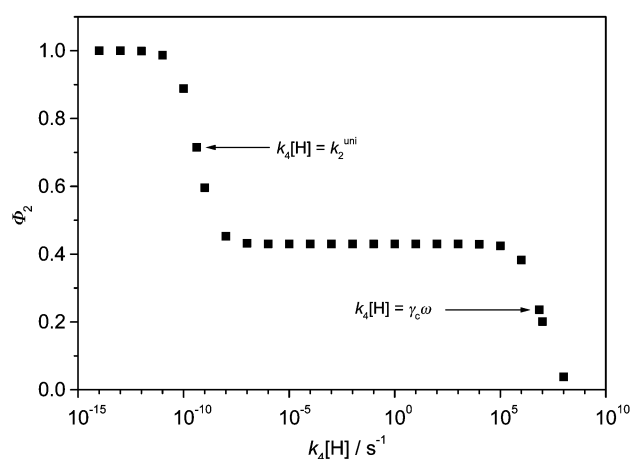


Fig. 2 Relative branching fraction of reaction (2) with respect to the pseudo-first order rate coefficient of the consecutive bimolecular reaction; for explanations see text.

inequality it can be estimated whether the branching ratio is affected by bimolecular reactions. Under certain conditions lower limits for rate coefficients of prospective scavenging reactions can also be estimated. If the time scale of a chemical activation experiment is distinctly larger than the reciprocal thermal rate coefficient for the unimolecular decomposition of the intermediate, $1/k^{\text{uni}}$, the final steady-state is normally reached.¹² If, nonetheless, relative yields below unity are measured for the unimolecular channels, a bimolecular reaction with a pseudo-first order rate coefficient larger than k^{uni} must exist. If the concentrations of possible reactants are known, lower limits for the bimolecular rate coefficients can be estimated. If these limits are not in accordance with known values or if the concentrations are so low that rate coefficients far above the collision limit would be required for an effective removal, the corresponding reaction can be excluded (see the discussions in ref. 16 on the capture of Criegee-intermediates in the ozonolysis of olefins under atmospheric conditions).

In the prototypical case studied here, the rates of collisional deactivation and thermal unimolecular decomposition differ by many orders of magnitude. This causes the different regimes to be well separated, which was one reason to choose $\text{s-C}_4\text{H}_9$ as an example (besides the fact that the input data were readily available). A remarkable influence of the scavenging reaction on the branching ratio, therefore, is restricted to quite uncommon conditions. For reactions with lower threshold energies of the unimolecular steps or chemical activation systems at lower pressures and/or higher temperatures, however, the intermediate range may be considerably smaller or even disappear. In these cases bimolecular consecutive reactions become important and, presumed the steady-state approximation holds, can be easily taken into account the way described here.

4. Conclusions

Chemically activated reactions are often described in terms of the branching between a unimolecular reaction and the collisional stabilization of the intermediate adduct (A) under steady-state conditions. The time range of this steady-state is approximately limited at the lower end by the characteristic time of collisional deactivation [$\text{ca. } (\gamma_c\omega)^{-1}$] and at the upper end by the average thermal lifetime of the adduct (k_{uni}^{-1}). If, however, a bimolecular reaction of the adduct $\text{A} + \text{B} \rightarrow \text{products}$ (rate coefficient k) exists, as is often the case in complex reaction systems, the intermediate steady-state remains an adequate assumption as long as $10k_{\text{uni}} < k[\text{B}] < 0.1\gamma_c\omega$. Depending on the temporal behaviour of the concentration [B], this may extend the range of applicability of the steady-state assumption to much longer times considerably facilitating the kinetic analysis.

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